

Fig. 5: One satellite along with its imaginary sphere

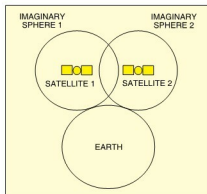


Fig. 6: Two satellites above the earth



Fig. 7: More than three satellites

is only one point—red in the program illustration—where the surfaces of the three virtual spheres meet on the surface of the earth.

The receiver generally looks to four or more satellites, for improved accuracy and precise altitude information (refer Fig. 7).

Software program

This software program demonstrates the concepts of GPS segments and 3-D trilateration. The coding is written in 'C' language and compiled using

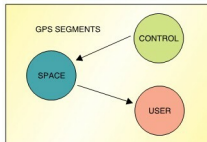


Fig. 8: Program output screenshot of GPS system

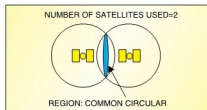


Fig. 9: Program output screenshot of two imaginary spheres

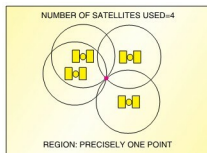


Fig. 10: Program output screenshot of more than three imaginary spheres

'Turbo C.' 'Turbo C' has a good collection of graphics libraries that has been utilised for graphics programming using basics of 'C'. The program uses various 'C' graphic files such as egavga.bgi, scri.chr, trip.chr, lcom.chr and tsr.chr.

GPS segments display. The first part of the program displays the three major segments (control, space and user) of the GPS system. The segments are displayed one by one with one-second delay between two consecutive segments (refer Fig. 8).

Demonstration of 3-D trilateration. The second part of the project demonstrates the principle of 3-D trilateration. First, it displays in a window screen the symbol of the satellite used in the program. Then the pro-

gram starts displaying the above-mentioned four cases sequentially one by one. In every display, corresponding to the number of the satellites considered, the region covered by the sphere is also indicated:

When one satellite is considered. A satellite with the imaginary sphere is displayed as shown in Fig. 5.

When two satellites are considered. In this case, two displays are shown: The first display shows two spheres corresponding to the two satellites (as shown in Fig. 6). The second display shows better understanding of the concept with a ring in blue colour (refer Fig. 9).

When three or four satellites are considered. In the third and fourth cases, the point of intersection of the spheres is made to blink in order to create a better presentation. The program output of more than three imaginary spheres is shown in Fig. 10.

Role of various functions

Initgraph(&gd,&gm,"c:\\turbo3\\bgi"). Initialises the graphics system by loading a graphics driver from the disk and then putting the system into graphics mode. It also resets all graphics settings like colour, palette and current position to their defaults. 'gd' specifies the graphics driver to be used and 'gm' specifies the initial graphics mode (unless 'gd' = DETECT). When gd = DETECT, the initgraph sets the graphics mode to the highest resolution available for the detected driver.

Closegraph(). Switches back the screen from graphics mode to text mode. It is called after getch() so that the screen does not clear until the user hits the key.

Void segments(). Displays the three major segments (control, space and user) of the GPS.

Void Satellite_symbol(). Displays the symbol of the satellite used in the program.

Void projectwork(). Demonstrates how the GPS receiver works using the principle of trilateration.